

CANDIDATE NAME: _____

INDEX NUMBER _____

CG _____



SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2017

BIOLOGY
Higher 2
9744

Tuesday
18 September 2017

2 hours

Additional materials:
Writing paper

READ THESE INSTRUCTIONS FIRST

Write your name and index number in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in all sections.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
Paper 1 (MCQ)	/30
Paper 2	
1	/15
2	/20
3	/15
4	/15
5	/18
6	/17
P2 Total	/100
Paper 3	/75
Paper 4	/55
TOTAL (100%)	

This question paper consists of 19 printed pages.

Answer **all questions** in the spaces provided.

Question 1

There have been many breakthroughs in stem cell research in recent years. It has been discovered that stem cells are involved in the replacement of worn-out cells and repair of damaged tissues. Further research is being conducted to better understand the mechanism involved in controlling the behaviour of stem cells in order to better manipulate them to treat various diseases and disorders.

- (a) State the type of stem cells involved in the replacement of worn-out cells and repair of damaged tissues, and describe the unique properties of this type of stem cells. [2]

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Stem cells undergo cell division to produce genetically identical daughter cells. **Fig. 1.1** shows two cells, each at a different stage of cell division.

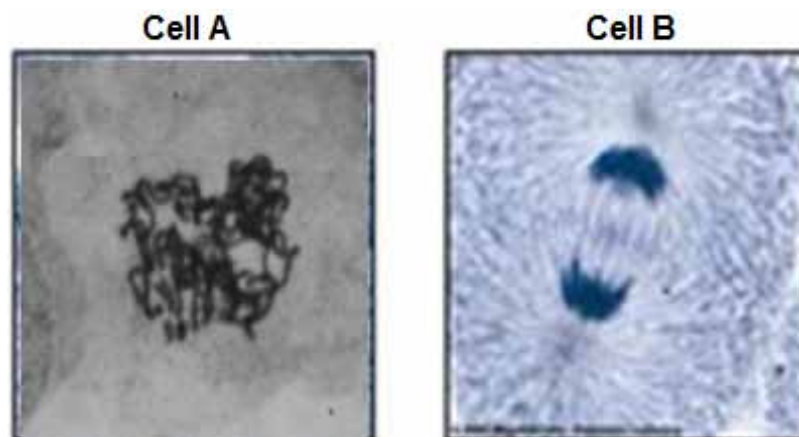


Fig. 1.1

- (b)(i) With reference to **Fig. 1.1**, state the stages of cell division in **Cell A** and **Cell B**. [1]

Cell A:

Cell B:

- (ii) The dysregulation of cell cycle can result in cancer. Outline the checkpoints that are present in normal cells to prevent this from occurring. [2]

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Fig. 1.2 shows information about the movement of chromatids in a cell that has just started metaphase of mitosis.

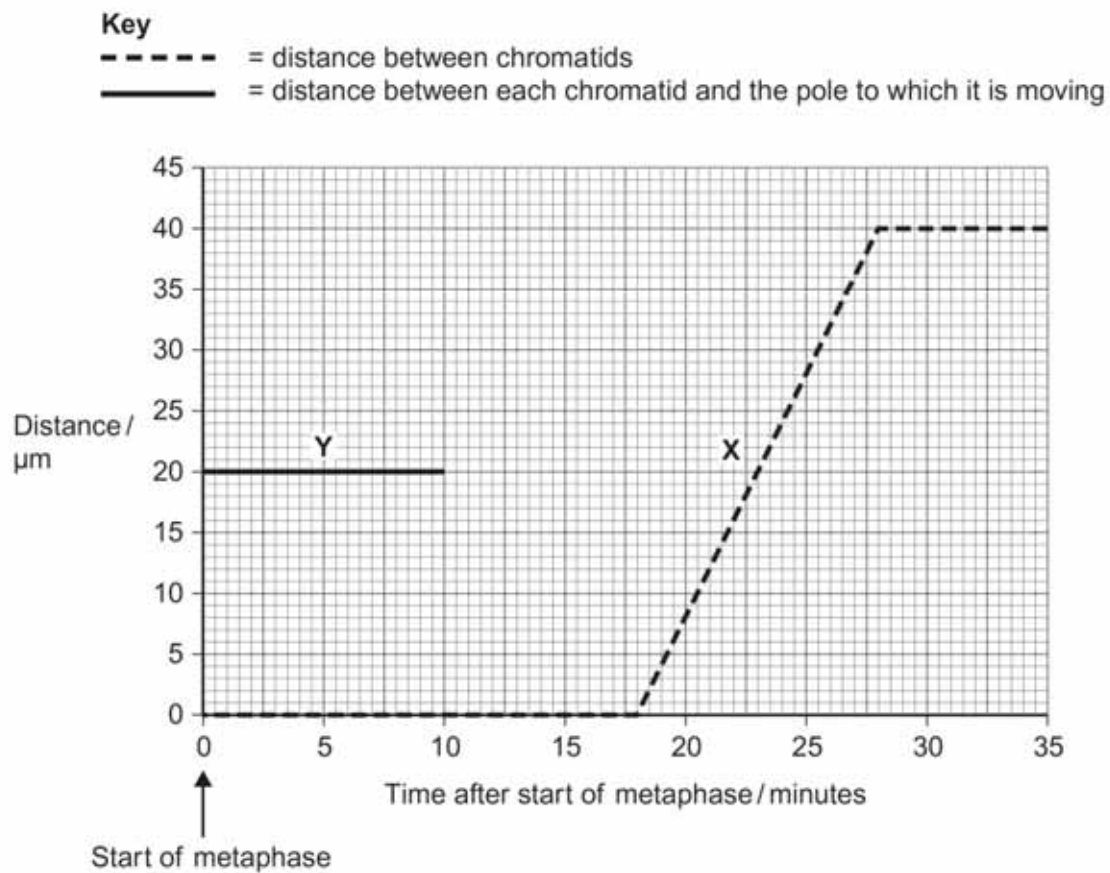


Fig. 1.2

- (c)(i) With reference to **Fig. 1.2**, state the duration of metaphase in the cell. [1]

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- (ii) Complete line Y on the graph. [1]

(iii) Account for your answer in **(c)(ii)**. [3]

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The movement of chromatids is dependent on spindle fibres, which are made up of many tubulin subunits. Spindle fibres are lengthened at one end during mitosis by the polymerisation of tubulin subunits through GTP hydrolysis.

A drug, eribulin, is known to prevent the polymerisation of the tubulin subunits.

(d)(i) Contrast between the structure of tubulin with that of DNA. [2]

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(ii) Suggest how eribulin work to prevent tubulin polymerisation and explain its effect on the behaviour of chromosomes in mitosis. [3]

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[Total: 15]

Question 2

The retrovirus, human immunodeficiency virus (HIV), and the influenza virus are two types of enveloped viruses. Both enter the human host cells by adsorption and penetration. **Fig. 2.1** shows the entry process of a HIV into a macrophage, which is a type of white blood cell.

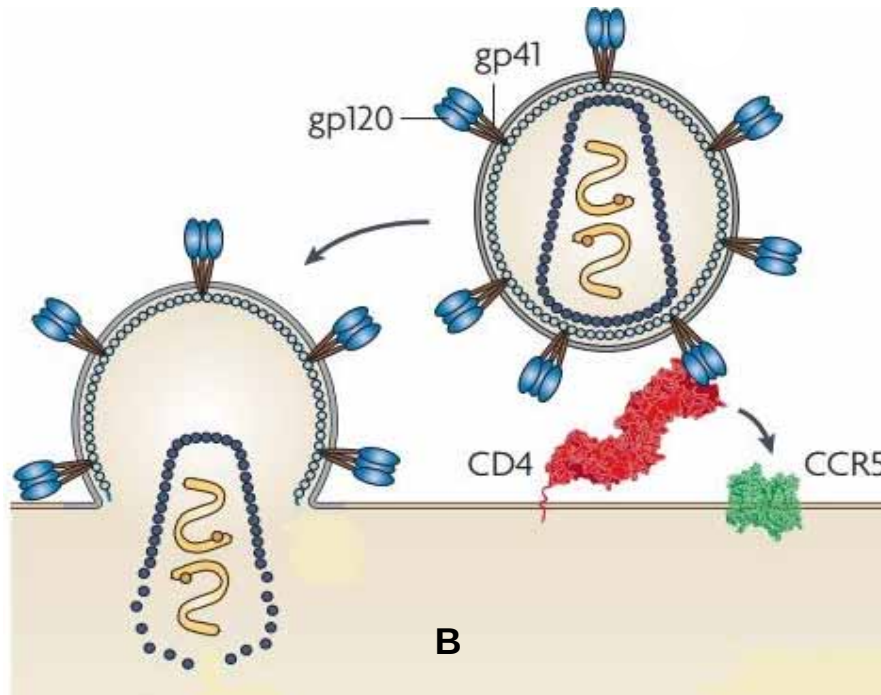


Fig. 2.1

(a)(i) State what is meant by *retrovirus*. [2]

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(ii) Compare the entry processes of the HIV and influenza virus into human host cells. [3]

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- (iii) Upon completion of the entry process, describe how the HIV genome is inherited by macrophage daughter cells. [3]

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- (iv) 'HIV-positive patients usually develop weak immunity.'

Explain what is meant by this statement. [4]

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Another example of an enveloped virus is the herpes simplex virus. Viral DNA enters the nucleus of the host cell via nuclear pores and *directs synthesis of viral RNA and DNA*. The virus is able to grow in non-dividing cells because its genome encodes enzymes such as viral DNA polymerase and thymidine kinase. Thymidine kinase is involved in synthesising deoxyribonucleotides required for DNA replication.

- (b)(i) Describe the normal function of nuclear pores. [2]

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- (ii) State the process that is involved in the [1]

Synthesis of viral RNA:

Synthesis of viral DNA:

(iii) Describe two ways in which the named processes in **(ii)** are different. [2]

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Acyclovir is an anti-viral drug which is used to treat herpes. It prevents the complete replication of viral DNA by viral DNA polymerase. The structure of acyclovir is shown below in **Fig. 2.2**.

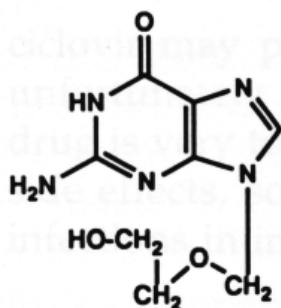


Fig. 2.2

When acyclovir enters the infected cell, it is first phosphorylated by thymidine kinase and subsequently becomes further phosphorylated by host cell kinases. When this phosphorylated acyclovir is incorporated into the newly-synthesised DNA strand, it prevents further elongation of the DNA strand.

(c)(i) With reference to **Fig. 2.2**, suggest how the incorporation of the phosphorylated acyclovir prevents further elongation of the DNA strand. [2]

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(ii) Some strains of herpes simplex virus are now resistant to acyclovir. Suggest how the virus has gained resistance to this anti-viral drug. [1]

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[Total: 20]

Question 3

In a particular variety of tomato plant, the allele for red fruit colour (**A**) is dominant over the allele for orange fruit (**a**) and the allele for green base when ripe (**B**) is dominant over the allele for no green base when ripe (**b**).

Two students, Faiz and Jacob crossed plant with red fruit and green base when ripe with pure bred plant with orange fruit and no green bases when ripe. The phenotypes of 50 offspring of each of Faiz's and Jacob's crosses were recorded and are shown in **Table 3.1**.

Table 3.1

	Phenotypes of offspring of test crosses			
	Red fruit with green base	Red fruit with no green base	Orange fruit with green base	Orange fruit with no green base
Faiz's cross	23	4	3	20
Jacob's cross	3	21	23	3

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- (a)(i)** With the aid of the table of probabilities as shown above, carry out a χ^2 test on the results of Faiz's cross and provide a brief explanation for the results obtained. [6]

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- (ii)** State the probability that the results of Faiz's cross depart significantly by chance from the expected ratio. [1]

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- (b)** Draw a genetic diagram to explain the results of Jacob's cross. [5]

- (c) Explain the difference in results between Faiz's and Jacob's cross when the parental genotypes are the same. [3]

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[Total: 15]

Question 4

Yeast haploid cells secrete α factor to signal mating, and respond by growing a mating projection towards a potential mate. Upon contact of the two partner cells, these fuse to form a diploid zygote.

Fig. 4.1 shows the α factor signaling pathway mediated by yeast G-protein coupled receptor, Ste2. The activation of the pathway induces the expression of *FUS1* gene which is required for yeast mating.

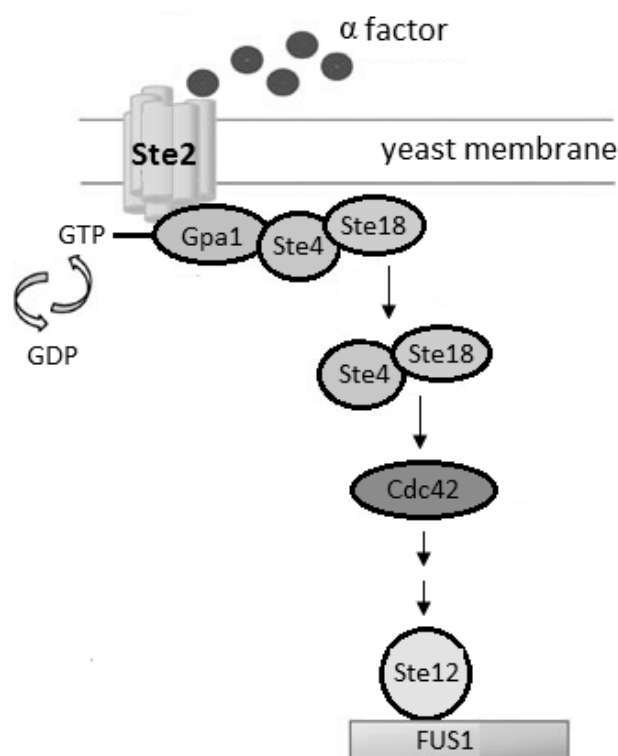


Fig. 4.1

(a)(i) With reference to **Fig. 4.1**, describe how α factor triggers the activation of Cdc42. [4]

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(ii) Briefly explain why α factor cannot enter the yeast cell directly. [2]

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(iii) Explain the possible role of Ste12 in the expression of FUS1 gene. [2]

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(iv) It has been observed that the binding of a protein Y to a region upstream of the promoter results in the mating projection to be produced at a rate higher than normal in the yeast cells.

Provide a reason for this observation. [3]

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Fig. 4.2 shows the Ste2 receptor on another yeast cell membrane.

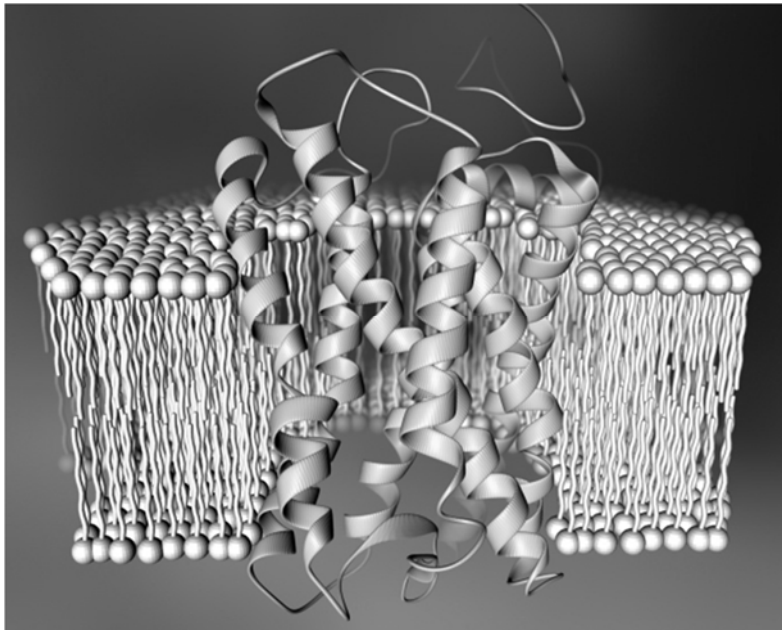


Fig. 4.2

(b)(i) Describe the structure of Ste2 receptor. [2]

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(ii) Explain how Ste2 receptor remains embedded in the yeast cell membrane. [2]

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[Total: 15]

Question 5

Arabidopsis thaliana is a small flowering plant native to Asia. A mutation in the gene coding for NADP oxidase results in plants with short root hairs. NADP oxidase is an enzyme that converts NADPH to NADP⁺.

Fig. 5.1 shows the root hairs in the two variant of *A. thaliana*.

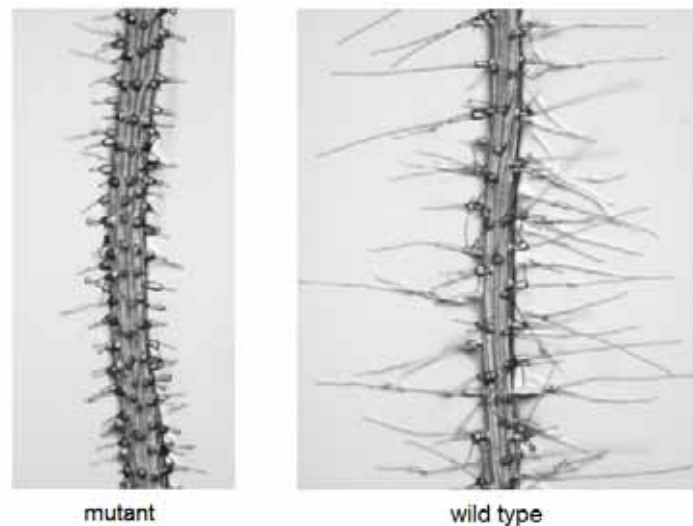


Fig. 5.1

(a) Explain the role of NADPH in photosynthesis. [2]

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In a separate experiment, activity of NADP oxidase in the tips of the root hair cells in wild type and mutant *A. thaliana* were measured at intervals. Changes in mean length of the root hair cell was also measured to track the rate of growth, which is known to be an energy-requiring process. Both sets of results are shown in **Fig. 5.2**.

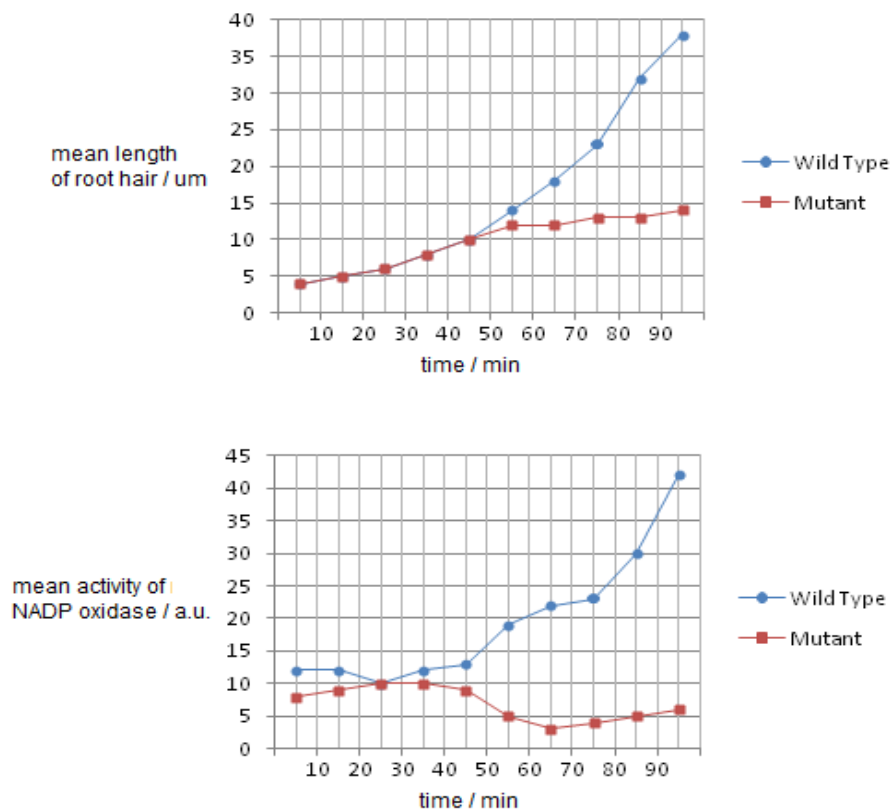


Fig. 5.2

- (b) Using your knowledge of photosynthesis, provide an explanation for the difference in the growth of root hairs in the two types of *A. thaliana*. [4]

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Rubisco is an enzyme required in the light-independent stage of photosynthesis. **Fig. 5.3** shows the effect of increasing temperature on the activity of two variations of Rubisco, **Rubisco C** and **Rubisco S**.

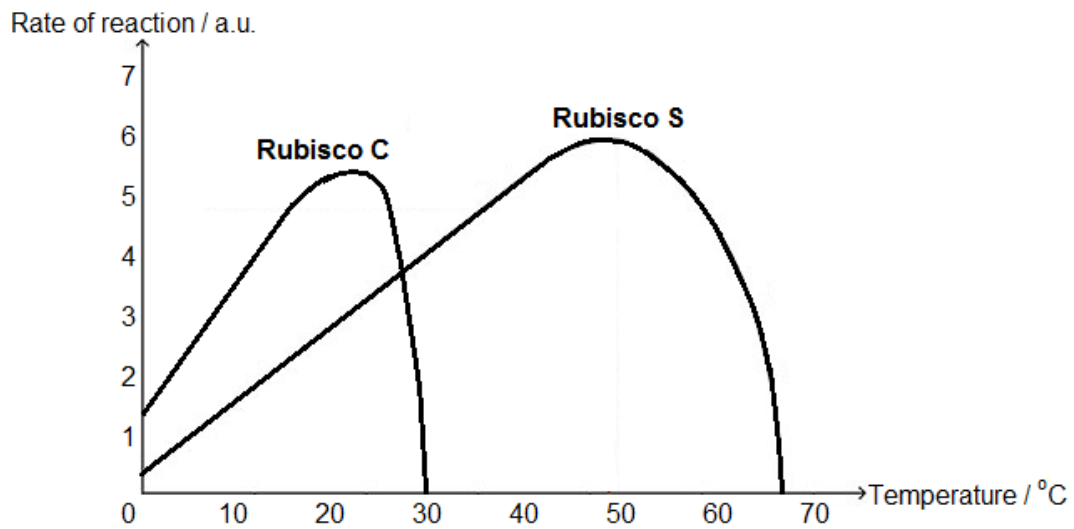


Fig. 5.3

- (c) With reference to **Fig. 5.3**, compare the effect of temperature on the two enzymes. [3]

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- (d) It is known that **Rubisco C** is obtained from a species of coniferous tree found in Canada, while **Rubisco S** is obtained from a species of cactus found in the Sahara Desert.

- (i) Explain how different alleles give rise to different Rubisco structure. [3]

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- (ii) It has been predicted that **Rubisco S** will be found in more plant species in view of climate change. Explain how Darwin's theory of evolution supports this observation. [4]

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- (iii) Suggest two other ways plants can adapt to the changing climate. [2]

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[Total: 18]

Question 6

The table below shows the amino acid differences in the cytochrome b protein between various vertebrates.

	Human	Elephant	Platypus	Ostrich	Starling	Crocodile	Lungfish	Coelacanth	Goldfish	Shark
Human		26	40	43	41	47	83	70	68	71
Elephant			45	45	48	50	84	72	63	74
Platypus				54	52	51	89	74	70	76
Ostrich					26	36	91	75	68	73
Starling						47	91	77	67	70
Crocodile							85	78	70	77
Lungfish								90	94	86
Coelacanth									83	78
Goldfish										88
Shark										

Fig 6.1 shows the phylogenetic tree based on differences between the cytochrome b proteins.

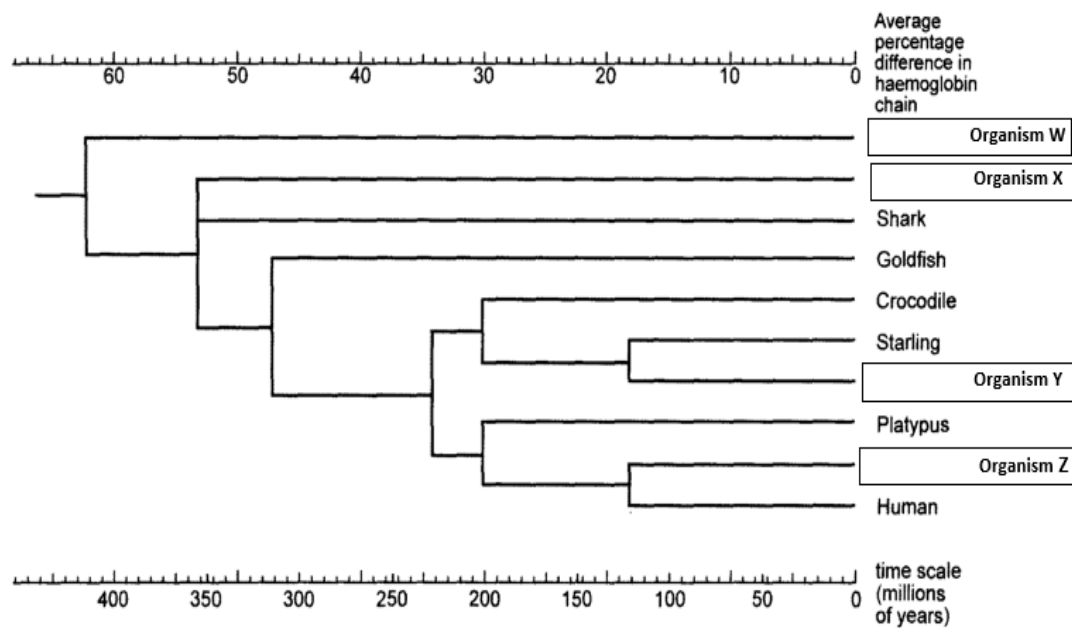


Fig. 6.1

(a) Using information from the table and **Fig. 6.1**, identify organisms **W** to **Z**. [2]

Organism **W**:

Organism **X**:

Organism **Y**:

Organism **Z**:

- (b) Distinguish between classification and phylogeny. [2]

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- (c) Explain how differences in amino acid sequences in the cytochrome b chain allow the establishment of the phylogenetic tree. [2]

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- (d) Suggest why homology still features prominently in evolutionary studies despite the advantages that molecular evidence can confer. [1]

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Brown adipocytes were one of the cells isolated for the above investigation. **Fig. 6.2** shows the schematic representation of a series of protein complexes found on the inner membrane of organelle **X** present in brown adipocytes.

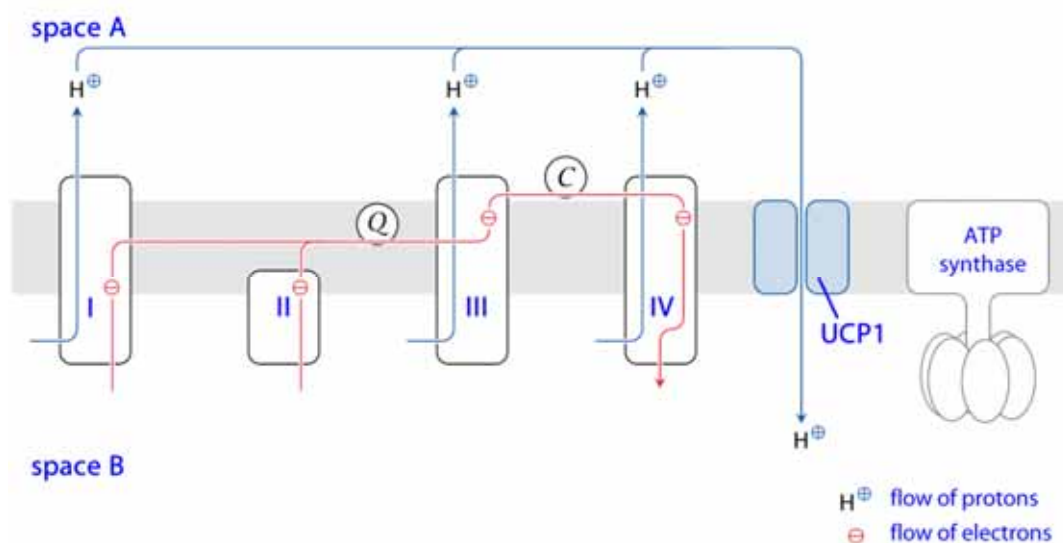


Fig. 6.2

(e)(i) State the identity of organelle **X**. [1]

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(ii) Outline how ATP is usually synthesised in the inner membrane of organelle **X**. [4]

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(f) Brown adipocytes contain a unique protein, UCP1, which is not found in organelle **X** in any other cell type.

Evaluate the impact of UCP1 on ATP synthesis and suggest the physiological significance of brown adipose tissue. [3]

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(g) In other cell types, NADH and FADH₂ are used to drive ATP synthesis by ATP synthase. Using relevant information from **Fig. 6.2**, suggest and explain why more ATP is produced from NADH. [2]

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[Total: 17]

END OF PAPER